QUESTION

For each S2 cell at level 13, how many sites are spatially connected to a location that contains that cell?

① IDENTIFY

1 search_all_classes()

"S2 cell"

sgul:S2Cell_Level13	**0.866**
sgul:GasNutrientLoss	0.001
sgul:YieldNutrientUptake	0.000

2 search_all_properties()

"spatially connected"

psss:connectedTo	**0.685**
sgul:sfContains	0.000
sgul:sfWithin	0.000

3 search_all_classes()

"Location"

ius:Location	**0.045**
osv:QuadMapFormat	0.001
oog:Geometry	0.000

4 search_all_classes()

"Site"

ius:Site	**0.033**
ius:GrowthStageManagement	0.000
ius:JournalArticle	0.000

🤔 *Identified the key classes (**S2Cell_Level13**, **Location**, **Site**) and the property **connectedTo**. But how are they connected?*

② EXPLORE

5 explore_property_usage()

psss:connectedTo

Domain	Range	Ex. Subject	Ex. Object	Count
S2Cell_Level13	**Location**	**s2.level13.983…**	**Location.ALAUEVS**	**★ 490**
ius:Site	Location	Site.BOZOISKY…	Location.CONUNRBS	10

6 explore_property_usage()

sgul:sfContains

Domain	Range	Ex. Subject	Ex. Object	Count
ius:Location	**ExperimentalUnit**	**Location.IAAM7071**	**EU.IAAM7071_101**	**★ 238**
ius:Site	**ExperimentalUnit**	**Site.GHG**	**EU.KYBGGHG_101**	**★ 237**
unknown	ExperimentalUnit	Site.KYGHG	EU.KYBGGHG_101	25

🤔 ***connectedTo** links S2Cell→Location (490 triples). **sfContains** links Location→ExperimentalUnit, not Site. Still missing the Location→Site link.*

7 search_all_properties()

"Location Site"

schema:workLocation	0.0036
psss:connectedTo	0.0031
rdf:type	0.0029
ius:hasCrop	0.0019
ius:cites	0.0018

… 15 more rows

❓ *Broad property search for Location↔Site link — no useful results. Falling back to entity-level exploration.*

② EXPLORE (cont.)

8 explore_entity_relations()

ius:Location.ALAUEVS

Property	Dir	Example neighbor
psss:connectedTo	**out**	**s2.level13.983…**
sgul:sfContains	out	EU.ALAUEVS_103…
sgul:sfWithin	out	adminRegion.USA.01
rdf:type	out	ius:Location
rdfs:label	out	GRACEnet
ius:fundedBy	out	FundingSource.7

9 explore_entity_relations()

ius:Site.GHG

Property	Dir	Example neighbor
psss:connectedTo	out	adminRegion.USA.21227
sgul:sfContains	**in**	**Location.KYBGGHG…**
sgul:sfContains	out	EU.KYBGGHG_101…
sgul:sfWithin	in	EU.KYBGGHG_101…
sgul:sfWithin	out	Location.KYBGGHG…
rdf:type	out	ius:Site
dct:identifier	out	GHG

🤔 *Step 9 reveals **sfContains** incoming from Location to Site — the missing link! Step 10 confirms from the Location side.*

10 explore_entity_relations()

ius:Location.KYBGGHG

Property	Dir	Example neighbor
sgul:sfContains	**out**	**EU.KYBGGHG_101…**
sgul:sfWithin	out	adminRegion.USA.21
rdf:type	out	ius:Location
rdfs:label	out	GHG
ius:hasTreatment	out	Treatment.KYBGGHG_1…
ius:cites	out	Publication.KYBGGHG.162…
ius:fundedBy	out	FundingSource.5 (NP212)
ius:startDate	out	2009-05-01
ius:endDate	out	2011-09-13

✅ *Full path resolved: **S2Cell** —connectedTo→ **Location** —sfContains→ **Site**. Ready to compose SPARQL.*

③ COMPOSE & VERIFY

11 execute_query()

```
SELECT ?cell
  (COUNT(DISTINCT ?site) AS ?cnt)
WHERE {
  ?cell a sgul:S2Cell_Level13 .
  ?cell psss:connectedTo ?loc .
  ?loc a ius:Location .
  ?loc sgul:sfContains ?site .
  ?site a ius:Site .
}
GROUP BY ?cell
ORDER BY DESC(?cnt)
LIMIT 20
```

Results

?cell	?cnt
s2.level13.9758960523…	3
s2.level13.5965661247…	3
s2.level13.5965661281…	3
s2.level13.5969398281…	2
s2.level13.5969530437…	2
⋮	
s2.level13.5965726243…	1
s2.level13.5965726247…	1
s2.level13.5969201523…	1
s2.level13.5969398243…	1
s2.level13.5969398247…	1

20 rows · ORDER BY DESC · F1 = 1.00

✅ *First execution returns 20 rows. Agent re-runs to confirm determinism. F1 = 1.00*